

IFS Function in Excel

1. What is IFS?

The IFS function checks several conditions in order and returns the result for the first condition that is TRUE. It solves problems where a value must be placed into one of many categories, such as grades, rating levels, or commission bands. A Data Analyst uses it to create labels and categories without writing IF formulas inside each other.

2. Syntax

=IFS(logical_test1, value_if_true1, [logical_test2, value_if_true2], ...)

Argument	Required?	What it accepts
logical_test1	Required	The first condition, such as B2>=90
value_if_true1	Required	The result if the first condition is TRUE: text in quotes, a number, or a calculation
logical_test2, value_if_true2	Optional	More condition and result pairs. Excel accepts up to 127 pairs.

Every condition must be followed by its own result. Excel checks the pairs from left to right and stops at the first TRUE condition. IFS is available in Excel 2019, Excel 2021, and Microsoft 365, but not in Excel 2016 or earlier.

3. Simple Example

Assign a performance level based on score:

	A	B	C
1	Student	Score	Performance
2	Aisha	92	Excellent
3	Rahim	78	Good
4	Nabila	65	Average
5	Tanvir	48	Needs Improvement

Formula in C2, copied down to C5:

=IFS(B2>=90,"Excellent", B2>=75,"Good", B2>=60,"Average", B2<60,"Needs Improvement")

For Rahim (78): 78>=90 is FALSE, then 78>=75 is TRUE, so Excel returns "Good" and stops. For Tanvir (48): the first three tests are FALSE and 48<60 is TRUE, so he gets "Needs Improvement".

4. Another Example

Label stock levels: 0 is Out of Stock, 20 or less is Low, anything else is In Stock:

	A	B	C
1	Product	Stock	Status
2	Mouse	0	Out of Stock
3	Keyboard	15	Low
4	Monitor	60	In Stock
5	Headset	20	Low

Formula in C2, copied down to C5:

=IFS(B2=0,"Out of Stock", B2<=20,"Low", TRUE,"In Stock")

For Mouse (0): the first test is TRUE, so Excel returns "Out of Stock". For Headset (20): the first test is FALSE and 20<=20 is TRUE, so it returns "Low". For Monitor (60): both tests are FALSE, so the final TRUE test returns "In Stock".

5. When to Use It

- Convert exam scores into grades (A, B, C, D, F)
- Rate employee performance as Excellent, Good, Average, or Needs Improvement
- Apply different sales commission rates based on the sales amount
- Label inventory as Out of Stock, Low, or In Stock
- Group customers into tiers based on total spending

6. Important Things to Remember

- IFS checks conditions from left to right and returns the result of the first TRUE condition. It ignores the rest.
- Order matters. With >= tests, go from the highest value to the lowest, or lower tests will catch high values first.
- Each condition needs its own result. A missing result causes an error.
- If no condition is TRUE, IFS returns #N/A. Use TRUE as the last condition to handle all remaining cases.
- Text results must be in quotation marks. Numbers and calculations do not need quotes.
- Use IF for one condition (two outcomes) and IFS for three or more outcomes.

7. Common Mistakes

Mistake: Putting the conditions in the wrong order, such as B2>=60 before B2>=90, so a score of 92 returns "Average". **Fix:** List >= conditions from highest to lowest.

Mistake: Getting #N/A because no condition was TRUE. **Fix:** Add TRUE, "result" as the last pair, or make sure your conditions cover every possible value.

Mistake: Leaving out a result or a comma, so the conditions and results do not form pairs. **Fix:** Read the formula as pairs: condition, result, condition, result.

Mistake: Forgetting quotation marks around text results. **Fix:** Write results like "Good" inside quotes.

8. Data Analyst Tip

After filling an IFS formula down a column, test the boundary values (for example 90, 75, and 60) to confirm each lands in the correct category. Then filter the new column to check for any #N/A results. This quickly shows gaps in your conditions.